

Menu Suggestion: An Effective Way to Improve Dietary Compliance in Peritoneal Dialysis Patients

Wei Chen, MD,^{*} Xin-hong Lu, RD,^{*} and Tao Wang, MD, PhD^{*,†}

Objective: Poor compliance with the dietary prescriptions is quite common in dialysis patients. We believe that most of the noncompliance is caused by the patient's poor understanding of the dietary prescription. Therefore, in the present study, we tried to investigate the role of menu suggestion in improving the patient's compliance with the dietary prescription.

Design and Setting: A longitudinal cohort study conducted at an outpatient dialysis clinic.

Patients: Seventy clinically stable patients on peritoneal dialysis were included in this prospective study during April 1, 2004, to November 31, 2004, in a single center. Patients who had significant cognitive impairment and thus did not understand the food contents during the training course were not eligible for enrollment.

Intervention: All the patients were randomly assigned to 1 of 2 groups. Group 1 patients received the traditional patient education method. Group 2 patients additionally received individualized menu suggestions based on their food preferences and education on how to exchange the foods at equivalent amounts according to the exchange list.

Main Outcome Measures: At present, there are no clear optimal dietary protein intake levels for peritoneal dialysis patients. Our experience is that a dietary protein intake level of 0.8 to 1.2 g/kg/d can maintain our patients in a good nutritional status. Thereafter, in this study we prescribed the dietary protein intake level at 0.8 to 1.2 g/kg/d and defined compliance as meeting this target protein intake level.

Results: There were 35 patients in each group. The compliance was 22.9% in group 1 and 57.1% in group 2 ($P < .05$).

Conclusions: Our study suggests that menu suggestion may be an effective way of improving the compliance with the diet in peritoneal dialysis patients. It improves the patient's understanding of the dietary prescription.

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DIETARY NUTRIENT INTAKE is thought to play a dominant role in maintaining nutritional status in peritoneal dialysis patients. Inadequate dietary nutrient intake will inevitably result in the development of protein-energy malnutrition, which contributes to poor clinical outcomes.¹⁻³ On the other hand, excessive dietary protein intake (DPI) will lead to metabolic acidosis, hyperphosphatemia, and accumulation of other ure-

mic toxins and will require more dialysis doses.⁴ It is thus important for the patients to adhere to the dietary prescriptions that are prescribed by the dietitian based on their nutrition requirements and dialysis adequacy. However, noncompliance is not uncommon in dialysis patients, especially compliance with the diet.^{5,6} In dialysis patients, noncompliance may lead to inappropriate dietary composition and protein intake, and therefore facilitate the occurrence of complications.

It has been increasingly recognized that noncompliance may be an important factor that determines the outcome of peritoneal dialysis therapy.⁷ Compliance is in general defined by the extent to which a person's and/or caregiver's behavior coincides with a health-promoting or therapeutic plan agreed on by the person (and/or family or community) and health care professionals.⁸ A number of variables have been identified and are thought to influence compliance. These variables

^{*}Division of Nephrology, First Hospital, Peking University, Beijing, China.

[†]Division of Nephrology, Third Hospital, Peking University, Beijing, China.

Address reprint requests to Tao Wang, MD, PhD, Division of Nephrology, Third Hospital, Peking University, 49 North Garden Road, Haidian District, Beijing 100083, China. E-mail: wangt@bjmu.edu.cn

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Table 1. Demographic Characteristics, Dietary Protein and Energy Intake, and Blood Chemistry in the Control Group and the Menu Suggestion Group (Mean $\pm$ SD)

	Control Group	Menu Suggestion Group
Number	35	35
Male:female	18:17	15:20
Age (y)	52.86 $\pm$ 14.86	57.57 $\pm$ 14.17
Number of diabetic patients	8	9
Education level (y)	12.72 $\pm$ 3.76	11.24 $\pm$ 3.26
Percent of patients who reached the targets	22.9% (8/35)	57.1% (20/35)*
Dietary protein intake at enrollment (g/kg/d)	0.82 $\pm$ 0.37	0.83 $\pm$ 0.38
Dietary protein intake at follow-up (g/kg/d)	0.84 $\pm$ 0.34	0.88 $\pm$ 0.25
Dietary energy intake at enrollment (kcal/kg/d)	25.6 $\pm$ 7.9	24.7 $\pm$ 8.3
Dietary energy intake at follow-up (kcal/kg/d)	24.2 $\pm$ 8.1	24.0 $\pm$ 6.8
Serum albumin at enrollment (g/L)	35.2 $\pm$ 3.5	35.5 $\pm$ 3.6
Serum albumin at follow-up (g/L)	35.9 $\pm$ 3.7	36.4 $\pm$ 2.8
Serum phosphate at enrollment (mmol/L)	1.67 $\pm$ 0.25	1.67 $\pm$ 0.23
Serum phosphate at follow-up (mmol/L)	1.68 $\pm$ 0.26	1.69 $\pm$ 0.15

* $P < .01$.

include age, gender, education, comorbidity, dialysis modality, time on dialysis, psychosocial adjustment to illness, health locus of control, various aspects of the patient-caregiver relationship, and the patient's knowledge about the disease and the regimens.⁶

Knowledge is one of the factors that is shown to be associated with compliance in previous studies, although conflicting data exist.⁷ Fran argued that to be compliant, individuals must understand what the regimen is and how it is to be followed.⁹ Education was found to be able to improve a patient's knowledge and was essential for influencing compliance behavior.¹⁰ We believe most of the noncompliance with the dietary prescription may be attributable to the patients' poor understanding of the dietary prescriptions because of the natural complexities of the food contents.

Therefore, in the present study, we investigated the effect of individualized menus on our patients' compliance with the diet by improving their understanding of dietary prescriptions.

Subjects

All new patients who started peritoneal dialysis in the peritoneal dialysis program of the First Hospital, Peking University, between April 1, 2004, and November 31, 2004, were asked to participate in the present study. Patients who had significant cognitive impairment and thus did not understand the food contents during the training course were not eligible for enrollment. The demographic characteristics of the study popula-

tion are shown in Table 1. Thirty-five patients were enrolled in each group. There were no significant differences in gender distribution, age, education level, and prevalence of diabetic patients between the 2 groups.

Method

Each patient received an intensive training course within 2 weeks after catheter implantation. Detailed information on food contents and appropriate diet were taught to the patients by an experienced dietician. Portion-sized food aids were used during the education, and the patients were educated to maintain a daily protein intake level of 0.8 to 1.2 g/kg/d. The patients were also taught how to correctly record their 3-day dietary intakes. After 3 months on peritoneal dialysis, we enrolled all patients who were clinically stable and were willing to participate in the study.

After enrollment, the dietician reviewed with each patient the food contents and the appropriate amount of protein intake, and all patients were then randomly assigned to 1 of 2 groups using random numbers. Patients in group 1 were educated by the traditional method described above and served as controls, whereas each patient in group 2 additionally received an individualized menu from the dietician based on his or her own food preference and an exchange list as a reference (Table 2).¹¹ In the exchange list, every food in one list contains an equivalent amount of protein. The patients were also taught how to correctly use their menu by referring to

Table 2. Exchange List for Chinese Foods

Food List	Serving Size (g)	Energy (kcal)	Protein (g)	Carbohydrate (g)	Fat (g)
Starchy foods*	25.0	90.0	2.0	20.0	–
Vegetables and fruits					
Vegetables†	500.0	90.0	5.0	17.0	–
Fruits‡	200.0	90.0	1.0	21.0	–
Meats and eggs					
Beans§	25.0	90.0	9.0	4.0	4.0
Milk¶	160.0	90.0	5.0	6.0	5.0
Meats and eggs	50.0	90.0	9.0	–	6.0
Oil					
Oil**	10.0	90.0	–	–	10.0
Nuts††	15.0	90.0	4.0	2.0	7.0

*Includes rice or flour (25 g); millet, corn powder, cereals, soda biscuit, steamed bread, or noodle (35 g); white potato (100 g); corn (200 g).

†Includes cabbage, cole, spinach, leek, fennel, celery, balsam pear, cucumber, eggplant, tomato, white gourd, or mushroom (500 g); white radish or green pepper (400 g); cushaw or cauliflower (350 g); carrot (200 g).

‡Includes pear, peach, apple, pomelo, orange, plum, apricot, grape, or pineapple (200 g); persimmon or lichee (150 g); strawberry (300 g); watermelon (500 g).

§Includes tofu (25 g); bean milk (400 g) (the ratio of bean and water is 1/8).

¶Includes milk (160 g); powdered milk (20 g); sheep milk (160 g); sugar-free yogurt (130 g).

||Includes lean pork (50 g); beef or mutton (50 g); chicken, duck meat, goose-, or medium-fat meat (25 g); hairtail or carp (80 g); cooked ham or sausage (20 g); egg (50 g).

**Includes peanut oil, balm balsam, colza oil, bean oil, lard, or butter (10 g).

††Includes walnut or pignut (15 g); sunflower seed (with shell) (25 g); watermelon seed (with shell) (40 g).

the exchange list. Portion-sized food aids were also used during the education.

All the patients were asked to record their diet history and to come back to the clinic 1 month later with their 3-day dietary records. The data were entered into the computer, and protein intakes were then calculated using a computer program (PD Information Management System, Peritoneal Dialysis Center, Peking University, Beijing, China). DPI was normalized to the individual's desirable body weight.

Each patient's compliance with the diet was arbitrarily defined as having met the target DPI of 0.8 to 1.2 g/kg/d. Any patient whose DPI was <0.8 g/kg/d or >1.2 g/kg/d was considered non-compliant.

Statistical analysis was performed using SPSS 11.0 for Windows software (SPSS Inc, Chicago, IL). All continuous variables were expressed as mean $\pm$ SD, and categorical variables were expressed as ratios. The comparisons between the 2 groups were performed using independent-sample *t* test or χ^2 test when applicable. $P < .05$ was considered statistically significant.

Results

The demographic characteristics of the patients are shown in Table 1. There were no significant

differences in demographic characteristics between the 2 groups of patients. All patients were rather stable at their clinical visits 1 month after their enrollment. Although there were no significant differences in serum albumin and phosphate levels between the 2 groups and between the enrollment and follow-up, the variation in the menu suggestion group at follow-up became much smaller. The profiles of DPI in the 2 groups are shown in Figure 1. There were 57.1% of the patients (20 of 35) in group 2 who met the target protein intake, whereas in group 1 (the control group), only 22.9% (8 of 35) met the target protein intake ($P < .01$). Therefore, patient com-

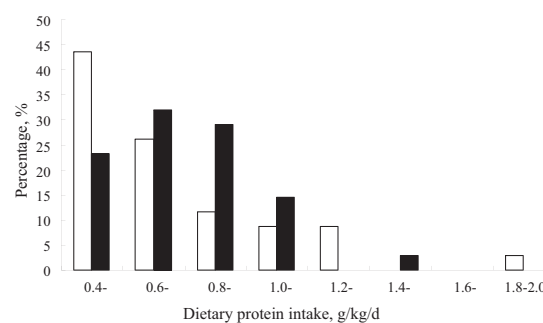


Figure 1. Distribution of dietary protein intake between the control group (blank) and the menu suggestion group (solid).

pliance in group 2 was significantly better than that in group 1.

Discussion

The present study found that even with intensive patient education on dietary intake, the number of patients who met the target protein intake was rather low. However, menu suggestion significantly improved the appropriateness of our patients' DPIs.

Nutritional support is an important part of the therapy of peritoneal dialysis, in which appropriate protein intake is definitely essential to ensure the patient a satisfactory nutritional condition.^{12,13} Either inadequate or excess protein intake is adverse to PD patients. Inadequate protein intake is one of the leading causes of development of malnutrition in PD patients.^{14,15} However, excess protein intake leads to inadequate dialysis, which may have a deleterious effect on patients.¹⁶ Nevertheless, it is still not clear what the optimal DPI level is for peritoneal dialysis patients. The National Kidney Foundation Clinical Practice Dialysis Outcomes Quality Initiative (DOQI) guidelines recommended a DPI of 1.2 to 1.3 g/kg/d for chronic peritoneal dialysis patients.¹⁷ However, recently more and more studies indicated that this value might be too high for PD patients, and the differences in geography and race should also be taken into account.^{18–20} Uribarri et al believed that the strong relationship between protein intake and phosphorus intake made it possible that this daily load of protein led to hyperphosphatemia in anuric patients with borderline dialysate clearance. Considering the inverse correlation between serum phosphorus and outcome among dialysis patients, probably mediated by an increase in the $\text{Ca} \times \text{P}$ product and the lack of evidence, theoretic or experimental, that such a high protein intake is by itself beneficial to dialysis patients, Uribarri used the general recommendation for normal subjects plus the usual excessive nitrogen loss during peritoneal dialysis, suggesting that protein intake between 0.9 and 1 g/kg/d most likely is an appropriate protein intake without exposing the patient to an excessive phosphorus load.²⁰ Similarly, Lim et al suggested that an adequate DPI for continuous ambulatory peritoneal dialysis (CAPD) patients is 1.0 to 1.1 g/kg/d.¹⁸ Sutton found that 59% of their CAPD patients were well nourished with a DPI of 0.9

g/kg/d.²¹ Moreover, a number of nitrogen balance studies have suggested that the average protein intake for maintaining protein balance was about 0.8 g/kg/d for most healthy adults²² and around 1.0 to 1.1 g/kg/d, or even as low as 0.7 g/kg/d, for chronic peritoneal dialysis patients.²³ It should be noted that an excessive DPI will lead to more difficulty in controlling hyperphosphatemia, metabolic acidosis, and accumulation of other uremic toxins and will require more dialysis doses.^{4,24} For example, for a 60-kg patient, a DPI of 1.2 g/kg/d would require a daily dialysis dose of 10 to 12 L, whereas a DPI of 0.9 g/kg/d would require a daily dialysis dose of 6 to 8 L to maintain the nitrogen balance and blood urea level at 25 mmol/L. It is our experience that a DPI level of 0.8 to 1.2 g/kg/d can maintain our patients in a good nutritional status while decreasing the dialysis cost, which is very important in our practice because of the financial limitations in our country. Therefore, in the present study, we prescribed the DPI level at 0.8 to 1.2 g/kg/d and defined compliance as meeting this target protein intake level.

The low compliance rate in group 1 was not surprising. We believed that traditional education methods might be too general for each individual. Although the patients might be fully willing to comply with the dietary prescription, considering the complexities of the food contents and the resultant difficulties in evaluating the correct protein amount of foods, it was possible that actually the patients complied with their prescriptions "wrongly". In other words, the patients thought they had fully understood and followed their dietary prescriptions, but they actually had not, and therefore when they applied the dietary prescription in the way they thought was correct, they did not receive the correct protein intake.

The significant improvement in compliance in group 2 after the menu suggestion was provided suggested that understanding the dietary prescription might be one of the key factors in improving patients' compliance with diets. Therefore, a patient's understanding was the bridge between dietary prescription and good compliance. Any method contributing to improving the patient's understanding will lead to better compliance. The results of our study showed that 57.1% of patients using the menu suggestion method met the dietary prescription, compared with 22.9% using the traditional method. We thought that

the individualized menu transformed the abstract concept of protein intake into visualized amounts of foods and was of course easier for the patients to understand. The patients clearly knew the protein contents in each food in their diets and then were aware of the whole protein intake that they received. As a result, it was much easier for the patients to control their protein intakes, and thereafter both inadequate and excess protein intake would less likely happen. It also made it possible for the patients to change different kinds of foods with equal amounts of protein at will. The menu suggestion with an additional exchange list served as a clearly presented reference to their diet planning. Patients simply selected some foods among their favorites on the menu, just like using a menu to order a meal in a restaurant. To sum it up, the menu suggestion method was more patient friendly, more easily understood, and more flexible for patient application than the traditional method.

It should be noted that even with the menu suggestion approach, a significant number of patients did not meet the target protein intake level, suggesting that better patient education methods should be developed. Repeated patient education with various menu suggestions is currently underway in our program to further improve the patient compliance with the dietary prescriptions.

In conclusion, our study showed that menu suggestion may be an effective method of improving the patient compliance with the diet by enhancing their understanding of dietary prescriptions.

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